



Ionic liquids as entrainers for the separation of azeotropic mixtures: Experimental measurements and COSMO-RS predictions

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HIGHLIGHTS

- Isobaric VLE measurements, for ethanol-water mixtures with ionic liquids.
- The ionic liquids examined increased the relative volatility of the ethanol-water mixture shifting the azeotrope.
- Theoretical study with the COSMO-RS model and two different approaches for treating ionic liquids.
- COSMO-RS is able to qualitatively correctly predict the experimental data.

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ABSTRACT

This work focuses on the evaluation of protic ionic liquids (PILs), prepared by direct neutralization of ethanolamine with different carboxylic acids, for being used as entrainers in the separation of ethanol-water azeotropic mixture. Isobaric vapor-liquid equilibria measurements, for ethanol-water mixtures with 2-hydroxyethylammonium formate and 2-hydroxyethylammonium butyrate, at initial concentrations of 5%, 10% and 15% mass fraction, have been carried out at 101.3 kPa. The results reveal that ILs increase the relative volatility of the ethanol-water mixture shifting the azeotrope towards higher ethanol concentrations, ultimately leading to complete azeotrope elimination. Intermolecular interactions between components of the binary alcohol-water and the ternary alcohol-water-IL mixtures were theoretically studied with the COSMO-RS model, using two different approaches to treat the IL: firstly, as a single compound and secondly, as a separate cation and anion. It is shown that COSMO-RS is able to qualitatively predict the experimental data, although significant quantitative deviations are observed.

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1. Introduction

Ionic liquids (ILs) have been widely investigated for a variety of applications in engineering, chemistry, biochemistry and many other areas. They are organic salts with low melting or glass transition temperatures consisting of organic cations and organic or inorganic anions. ILs, except for their wide liquid range, possess unique characteristics such as negligible vapor pressure, good solvent capacity for polar and nonpolar components, high thermal and chemical stability. These remarkable properties make them possible solvents for use in safer and more environmentally friendly processes. Of course, toxicological research studies have demonstrated that commonly used ILs have a certain level of toxicity, so special attention should be paid for specific IL applications. It is worth mentioning that the properties of ionic liquids can be

tuned to a great extent by the appropriate combination of cations and anions. This tunability, along with the above-mentioned number of properties, led to the invention of the term designer solvents to emphasize the possibility of designing an IL to match the requirements of a specific application. The characteristics of ILs render them attractive not only for their use as solvents, but also for alternative roles in a broad range of varied applications such as organic synthesis, electrochemistry, liquid phase extraction, catalysis for clean technology, polymerization processes etc. Among the several applications foreseeable for ILs in the chemical industry, there has been considerable interest in the potential of ILs for separation processes as extraction media. In the last years, ILs are under intensive investigation to determine their potential as replacement solvents for extractive distillation (Boli et al., 2018; Gjineci et al., 2016; Tsanas et al., 2014; Calvar et al., 2010; Calvar et al., 2007; Calvar et al., 2006; Calvar et al., 2008; Calvar et al., 2009; Ge et al., 2008; Geng et al., 2010; Orchillés et al., 2010; Rodríguez et al., 2015; Zhao et al., 2006). Some of the main

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advantages related to the replacement of conventional entrainers for extractive distillation, especially for separation of azeotropes, with ILs are: (1) they show a good solvent capacity for both polar and nonpolar compounds to be separated, (2) no impurities of ILs are present in the distillate stream due to their negligible vapor pressure; (3) tuning of their physical properties, e.g. viscosity, heat capacity etc., by appropriate combination of cations and anions; (4) high thermal and chemical stability and (5) low melting points, which means that devices, such as heat exchangers, can perform well at very low temperatures. However, the design of such processes requires the availability of phase equilibrium data of systems containing ILs.

Although the number of potential combinations of cations and anions to synthesize an ionic liquid is almost unlimited, which can be considered as an import advantage, it can be a drawback since experimental data concerning their thermophysical properties and phase equilibrium are very scarce due to the fact that they are costly and time consuming. Consequently, the use of predictive tools to estimate these properties of pure ILs and their mixtures is of great importance. In this regard, the COnductor-like Screening MOdel for Real Solvents (COSMO-RS) (Klamt and Eckert, 2000), is used in this work as a predictive tool to describe the vapor-liquid equilibria (VLE) values of mixtures containing ILs. There exists a considerable body of literature on COSMO-RS use for the description of phase equilibria of mixtures containing ILs with the majority of it dealing with the application of COSMO-RS for ILs screening. Diedenhofen *et al.* (Diedenhofen *et al.*, 2003) reported the calculated activity coefficients at infinite dilutions for 38 compounds in the ILs: [Bmpy] [BF₄], [Emim][NTf₂], and [Em₂im][NTf₂] and found that the root mean square deviations for $\ln \gamma_i^{inf}$ were not bigger than 0.7 ln-units. COSMO-RS predictions of infinite dilution activity coefficients of a series of hydrocarbons and alcohols on three phosphonium-based ILs, have been also reported by Banerjee *et al.* (Banerjee and Khanna, 2006) and they were found to match closely with experimental values. Anantharaj and Banerjee (Anantharaj and Banerjee, 2011) reported the COSMO-RS infinite dilution activity coefficients predictions of thiophene in 8 imidazolium based ILs with an 11% average root mean square deviation (RMSD) from experimental data. In the light of the reported studies pertaining to the COSMO-RS prediction of infinite dilution activity coefficients of different compounds in ILs (Gutiérrez *et al.*, 2012; Kato and Gmehling, 2005; Khan *et al.*, 2016; Kurnia *et al.*, 2014; Kurnia *et al.*, 2014; Lei *et al.*, 2006; Paduszynski, 2017; Wlazło *et al.*, 2016), it can be concluded that, generally, COSMO-RS can provide reliable qualitative results with reasonable quantitative deviations from experimental data.

Several other studies refer to COSMO-RS predictions of liquid-liquid equilibrium (LLE) for systems containing ILs, mostly of them focusing on binary mixtures. The majority of the research deals with LLE predictions of IL-hydrocarbons mixtures (Banerjee *et al.*, 2008; Domańska *et al.*, 2006; Ferreira *et al.*, 2011; Ferreira *et al.*, 2012; Gómez *et al.*, 2014; Kumar and Banerjee, 2009; Manohar *et al.*, 2013; Potdar *et al.*, 2012; Varma *et al.*, 2011) where COSMO-RS were found to be in good agreement with experimental data. However, when it comes to polar systems COSMO-RS does not accurately predict the polarity differences of the compounds (Gómez *et al.*, 2014; Domańska *et al.*, 2006; Freire *et al.*, 2007; Oliveira *et al.*, 2012; Wu *et al.*, 2003).

Banerjee and coworkers (Banerjee *et al.*, 2006) used COSMO-RS to predict binary vapor-liquid equilibrium (VLE) of systems containing a set of 13 IL systems ([C_xmim][NTf₂](x = 1, 2, 4), [C₁mim][Me₂PO₄] with various solvents. They found that the deviations lie in the expected range of accuracy for a priori predictions with the RMSD for bubble point pressure prediction being about 6%. Further-

more, several binary systems of imidazolium and pyridinium-based ILs with common alcohols such as ethanol, propanol, 2-propanol and methanol have been considered for VLE prediction in the study carried out by Freire *et al.* (2007). The temperature dependence of the liquid phase non-ideality and the positive deviation from Raoult's law with decreasing alkyl chain length has been described adequately with COSMO-RS. An overestimation of the deviations of Raoult's law appeared for very short chain alcohols, such as methanol. VLE predictions by COSMO-RS for aqueous IL binary mixtures have been also reported by the same group (Freire *et al.*, 2008). A reasonable qualitative prediction was obtained for the various ILs, with the deviations from experimental data increasing with the increase of the hydrophilic character of ILs' anion. Several other studies focus on the COSMO-RS approach for ILs evaluation as potential entrainers in the separation of azeotropic mixtures (Dhanalakshmi *et al.*, 2013; Janakey Devi *et al.*, 2018; Janakey Devi *et al.*, 2018; Li *et al.*, 2012; Verma and Banerjee, 2010). From a synopsis of the available literature, it can be stated that COSMO-RS is an efficient tool for a priori predictions of systems involving ILs, which may be of considerable value for the selection of suitable ILs for specific applications.

The present study is a part of our ongoing work on experimental vapor-liquid equilibria measurements aiming to identify the effect of ILs on azeotropic mixtures (Boli *et al.*, 2018; Gjineci *et al.*, 2016; Tsanas *et al.*, 2014). More specifically, in this work, the ability of two protic ILs, namely 2-hydroxyethylammonium formate (2HEAF) and 2-hydroxyethylammonium butyrate (2HEAB), in facilitating the separation of azeotropic ethanol-water mixture is experimentally investigated through isobaric vapor-liquid equilibria measurements at 101.3 kPa for different IL initial concentration (5%, 10% and 15% in mass fraction). Furthermore, the performance of COSMO-RS in the prediction of vapor-liquid phase equilibrium of systems containing ionic liquids is investigated. COSMO-RS is used to predict the VLE of mixture containing alcohols (ethanol and isopropanol), water and ILs (2-hydroxyethylammonium formate, 2-hydroxyethylammonium acetate and 2-hydroxyethylammonium butyrate). In addition, the molecular interactions that govern the phase equilibria of the studied mixtures are investigated and discussed. Two different approaches are considered for describing the ILs and their influence in VLE mixtures: in the first case, IL is built from a separate cation and a separate anion and in the second case, as a single compound.

2. Experimental section

2.1. Materials and methods

Ethanol, ethanolamine, water, Aqualine™ Titrant 5 and the solvent for the Karl-Fischer titration were purchased from Fisher Scientific. Formic acid was purchased from Chem-Lab and butyric acid from FERAK. All reagents were used without additional purification. The source and purities of the chemicals used are listed in Table 1.

The two ionic liquids studied in this work, 2HEAF and 2HEAB, have been synthesized in our laboratory following the procedures described in our previous work (Boli *et al.*, 2018; Gjineci *et al.*, 2016). Equimolar amounts of ethanolamine and acid were used for each synthesized IL.

¹H NMR spectra (300 MHz) for the characterization of ionic liquids were recorded on a Varian Gemini 2000 300 MHz spectrometer and can be found in the Supporting Information (Figs. S1 and S2). More information about the characterization of PILs used in this work can be found in (Tzani *et al.*, 2016).

Table 1
Materials description.

Chemical name	Source	CAS RN	Purification method	Mole fraction purity	Water mass fraction	Analysis method
Ethanol	Fisher Scientific	64-17-5	None	0.999	0.010	GC ^a
Water	Fisher Scientific	7732-18-5	None	0.990	–	GC ^a
Aqualine™ Titrant 5 and solvent for Karl-Fischer titration	Fisher Scientific	–	None	–	–	–
Ethanolamine	Fisher Scientific	141-43-5	None	0.990	0.005	GC ^a
Formic acid	Chem-Lab	64-18-6	None	0.980	–	GC ^a
Butyric acid	FERAK	107-92-6	None	0.990	0.0050	GC ^a
2-hydroxyethylammonium formate, (2HEAF)	synthesis	53226-35-0	In vacuo evaporation	0.990	0.0054	¹ H/ ¹³ C NMR ^a spectroscopy and KF ^c
2-hydroxyethylammonium butyrate, (2HEAB)	synthesis	56409-18-8	In vacuo evaporation	0.990	0.0048	¹ H/ ¹³ C NMR ^a spectroscopy and KF ^c

^a The analysis method for mass fraction purity was provided by the suppliers. GC: gas chromatography.

^b Nuclear Magnetic Resonance.

^c Karl Fischer titration.

2.2. Vapor-Liquid equilibria measurements

Vapor-liquid equilibria measurements were performed in the FISCHER LABODEST® VLE 602 (i-Fischer® Engineering GmbH) ebulliometer. The apparatus is a dynamic recirculation still where intimate contact between liquid and vapor phase takes place and it is composed of a heating and equilibrium cell and two condensers. A detailed description is given by Voutsas *et al.* (Voutsas *et al.*, 2011). All the experiments were carried out under adiabatic and isobaric conditions until equilibrium was reached. The equilibrium temperature was measured by means of a PT100 temperature sensor with an uncertainty of ± 0.01 K. As far as pressure is concerned, this has been measured with an accuracy of ± 0.01 kPa by a digital manometer. Condensed liquid and vapor phases were sampled and their water content was determined using a TitroLine Karl-Fischer Titrator. The entrainer's mass fraction of the liquid phase sample was determined gravimetrically by evaporating the volatile components. Molar fractions were determined with an accuracy of ± 0.001 . The method used in this work for the VLE measurements has been previously validated by measuring the binary ethanol-water mixture at 101.3 kPa (Tsanas *et al.*, 2014). Measured data have been found to be thermodynamically consistent proving the reliability of the method.

3. COSMO-RS model

The Conductor-like Screening Model for Real Solvents (COSMO-RS), developed by Klamt and co-workers (Klamt and Eckert, 2000), is an efficient method for the prediction of the thermodynamic properties of pure and mixed fluids using only the structural information of the molecules. The detailed description of the COSMO-RS model is beyond the scope of this work and only basic features will be presented here for the sake of discussion.

In the COSMO-RS theory, the solute is considered as an ensemble of closely packed ideally screened molecules. Molecular interactions are essentially considered as pairwise interactions of molecular surface segments, each of them having an average ideal screening charge density, σ , which is a valuable descriptor for the local polarity of molecular surface.

One main advantage of COSMO-RS is that it provides the contributions of the different intermolecular interactions to the total interaction energies, E_{int} , allowing thus the understanding of the phase behavior from a molecular point of view. The interaction energies include misfit energy (E_{misfit}), hydrogen bonding (E_{hb}) and van der Waals energy (E_{vdW}), calculated as following:

$$E_{misfit} = a_{eff} e_{misfit}(\sigma, \sigma') = a_{eff} \frac{\alpha'}{2} (\sigma + \sigma')^2 \quad (1)$$

$$E_{hb} = a_{eff} e_{hb}(\sigma, \sigma') = a_{eff} c_{hb} \min\{0; \sigma_{donor} + \sigma_{hb}\} \max\{0; \sigma_{acceptor} - \sigma_{hb}\} \quad (2)$$

$$E_{vdW} = a_{eff} e_{vdW}(\sigma, \sigma') = a_{eff} (\tau_{vdW} + \tau'_{vdW}) \quad (3)$$

In the above equations, σ and σ' are the screening charge densities of two different segments; a_{eff} , α' , c_{hb} , and σ_{hb} are the effective contact surface area, the misfit energy constant, the hydrogen bonding coefficient, and the cutoff of hydrogen bonding, respectively; σ_{donor} and $\sigma_{acceptor}$ represent the screening charge densities of hydrogen bond donor and acceptor segments, respectively; and τ_{vdW} is the element-specific vdW interaction parameter.

For COSMO-RS calculation, the probability distribution of finding a surface segment with a specific screening charge density, namely σ -profile, is firstly derived from quantum chemical calculations. Based on the obtained σ -profile, the affinity of a solvent for a molecular surface of polarity σ can be investigated by calculating its σ -potential, $\mu_S(\sigma)$:

$$\mu_S(\sigma) = -kT \ln \left[\int p_{S(\sigma')} \exp \left(-\frac{E_{int}(\sigma, \sigma') - \mu_S(\sigma')}{kT} \right) d\sigma' \right] \quad (4)$$

where k is the Boltzmann constant and T is the system temperature.

The component chemical potential can be quantified from the statistical thermodynamics treatment of molecular interactions and is given by the following equation:

$$\mu_S^{X_i} = \mu_{CS}^{X_i} + \int p^{X_i}(\sigma) \mu_S(\sigma) d\sigma \quad (5)$$

where $\mu_{CS}^{X_i}$ is an additional area and volume dependent combinatorial term, which is added to account for the size and shape differences of the molecules in the system.

The activity coefficient of compound X_i in the solvent S , $\gamma_S^{X_i}$, is calculated as following:

$$\gamma_S^{X_i} = \exp \left(\frac{\mu_S^{X_i} - \mu_{X_i}^{X_i}}{RT} \right) \quad (6)$$

4. Computational details

In this work, COSMO-RS calculations have been performed following a two-step procedure. Firstly, the quantum chemical

COSMO calculations have been carried out for the molecular species involved so as to obtain the COSMO energy and the screening charge density, σ . Then, COSMO-RS calculations have been performed in order to calculate the chemical potential of the components involved.

The quantum chemical COSMO calculations have been carried out on the density functional theory (DFT) level, using the BP functional with RI (resolution of identity) approximation and a triple- ζ valence polarized basis set (TZVP). All structures have been fully optimized. All calculations have been carried out with the TURBO-MOLE program package. COSMO-RS calculations have been performed using the COSMOtherm program, (Version C2.1 Release 01.11) and the BP_TZVP_C21_0111 parameterization has been used.

Ionic liquids in the COSMO-RS framework can be described either as a mixture of distinct ions [C + A] or as an ion-pair single compound [CA]. In the first case, the cation and anion are optimized independently and then the IL is described as two charged species, with highly polarized functional groups, without introducing any cation-anion interaction. In the second case, cation and anion are optimized as a single component.

For a given pressure, the COSMOtherm calculates the isobaric VLE using the following equation:

$$p^{\text{tot}} = \sum_i p_i^s x_i \gamma_i, i = 1, 2, \dots, NC \quad (7)$$

where p^{tot} is the total pressure of the system and p_i^s is the vapor pressure of the pure component i . Also, x_i is the liquid phase mole fraction of the component i , γ_i is its liquid phase activity coefficient that is predicted by COSMO-RS, and NC is the number of components in the mixture.

Ideal behavior of the vapor phase is assumed, since the investigated systems here exist at low pressure. The vapor phase mole fractions y_i are obtained by Eq (8).

$$y_i = \frac{p_i^s x_i \gamma_i}{p^{\text{tot}}} \quad (8)$$

The vapor pressure of components was calculated by using the Antoine equation.

5. Results and discussion

5.1. Experimental VLE data for the ethanol-water-IL ternary systems

The VLE of ethanol-water-IL ternary systems was measured at atmospheric pressure for three initial IL (entrainer) concentrations in the feed solution: $\approx 5\%$, $\approx 10\%$ and $\approx 15\%$ mass fraction. The experimental data is tabulated in Tables 2 and 3. The results show that the equilibrium temperatures of the aqueous ethanol mixtures increase due to the addition of the non-volatile entrainer; the more entrainer is introduced into the system the higher the equilibrium temperatures are. Moreover, the relative volatility of the ethanol-water system increases as the mass fraction of the entrainer increases, shifting the azeotropic point to higher ethanol concentrations. The relative volatilities decrease with increasing the alkyl chain length of the IL, i.e. 2HEAF exhibits a greater salting-out effect on ethanol as compared to 2HEAB. More information about the topology of azeotropic mixtures can be found in (Jaubert et al., 2014). Fig. 1 presents a graphical illustration of the relative volatilities with respect to ethanol liquid mole fraction in IL-free basis.

The ethanol-water system is by far one of the most investigated azeotropic systems due to its industrial importance. A number of studies in the open literature have examined the use of ILs as entrainers for the separation of ethanol-water azeotropic mixtures,

but these are limited to imidazolium-based ILs. Experimental isobaric VLE data on ethanol-water-ILs ternary mixture have been collected and the effect of ILs on the relative volatility of the system has been compared at approximately 0.95 M fraction of ethanol (in free-basis) and 0.1 mass fraction of IL (Fig. 2). Except for the data shown in Fig. 2, there is a number of experimental data, which cannot be directly used for comparison due to the different entrainer's mass concentration involved. For instance, Li et al. (Li et al., 2012) have reported vapor pressures of ethanol-water-2HEAF mixtures at fixed IL mass fraction of 0.30. Although a direct comparison is not possible, since Li et al. have measured at higher IL mass fraction, the trend of boiling temperatures with IL concentration show a qualitative agreement between the data of Li et al. and our data. In addition, Zhao et al. (Zhao et al., 2006) except for [Bmim][Br], which is shown in Fig. 2, have also investigated [Mmim][DMP], [Emim][DEP], [Bmim][Cl] and [Bmim][PF₆] as entrainers for the separation of ethanol-water. From the relative volatilities reported, it is concluded that the effect of [Emim][DEP] and [Bmim][PF₆] is similar to that of 2HEAH, while [Mmim][DMP] shows the same behavior as [Emim][N(CN)₂]. On the other hand, [Hmim][Cl] has shown the smallest effect on the separation of ethanol-water system as reported by Zhang et al. (Zhang and Conrad Zhang, 2002). Although [Cl]⁻ is one of the most promising anions in the separation of this mixture, the cation's chain length is responsible for the low efficiency of the separation (Pereiro et al., 2012). VLE experimental data for the ethanol-water-[Emim][TfO] mixture provided by Orchillés et al. (Orchillés et al., 2010) should be also taken into consideration. The effect of the [Emim][TfO] on the VLE of the ethanol-water system is slightly smaller than that of [Emim][Cl], [Bmim][Cl], [Emim][Br] and [Bmim][Br] as stated by the authors. Consequently, it can be concluded that 2HEAF, 2HEAA and [Emim][OAc] are the most promising entrainers for the separation of ethanol-water azeotropic mixture.

5.2. COSMO-RS predictions

5.2.1. VLE prediction of binary alcohol-water mixtures

At first, it is necessary to check the accuracy of the COSMO-RS model in the prediction of the phase behavior of binary mixtures consisting of alcohols (ethanol and isopropanol) and water. VLE predictions of binary alcohol-water systems have been initially performed using the conformers of these compounds from the COSMOtherm database. For both alcohols, a weighted mixture of two conformers is considered in the database. Based on the experimental data available in the literature (Gmehling, 1977), both alcohols form azeotropes with water, and the minimum experimental azeotropic temperature and alcohol liquid-phase composition at 101.3 kPa are 360.8 K and 0.894 mol/mol for the ethanol-water mixture, and 353.1 K and 0.684 mol/mol for the isopropanol-water mixture. The COSMO-RS predictions using the alcohol conformers from the COSMOtherm database shown in Fig. 3, are not in good agreement with experimental data. Furthermore, in the case of the ethanol-water mixture, COSMO-RS does not predict any azeotropic point. For isopropanol-water mixture, the predicted azeotrope shows a deviation of $\Delta T = 1.8$ K in temperature and $\Delta x = 0.1$ mol/mol in isopropanol molar fraction from the experimental data. This finding is consistent with similar ones presented in literature. For example, Banerjee et al. (Banerjee et al., 2006) concluded that azeotropic systems that contain alcohols show high deviations from the experimental data, which is also confirmed by Xue et al. (Xue et al., 2012). For the ethanol-water system, although COSMO-RS describes qualitatively correctly the temperature dependence of activity coefficient at infinite dilution of ethanol (γ^∞), it underestimates the activity coefficients by a factor of ~ 0.2 log units (Klamt, 2012).

Table 2Isobaric VLE data^a at 101.3 kPa for ethanol (1)-water (2)-2HEAF (3).

% $w_{3,initial}$	T (K)	x_1	x_3	y_1	γ_1	γ_2	α_{12}
≈5.2	352.90	0.649	0.019	0.7463	1.089	1.617	1.50
	352.35	0.717	0.018	0.7879	1.062	1.731	1.37
	351.73	0.779	0.024	0.8351	1.061	1.854	1.28
	351.70	0.822	0.022	0.8668	1.045	1.893	1.24
	351.43	0.869	0.022	0.9041	1.042	1.953	1.19
	351.40	0.915	0.022	0.9408	1.031	2.107	1.09
	351.48	0.923	0.020	0.9464	1.025	2.139	1.08
	351.35	0.943	0.023	0.9669	1.030	2.186	1.07
≈10.3	352.02	0.735	0.039	0.8290	1.104	1.104	1.49
	351.90	0.785	0.045	0.8680	1.087	1.087	1.44
	351.85	0.807	0.042	0.8820	1.077	1.077	1.40
	351.71	0.815	0.044	0.8890	1.080	1.085	1.38
	351.58	0.866	0.045	0.9260	1.064	1.064	1.29
	351.70	0.893	0.046	0.9480	1.052	1.052	1.26
	351.71	0.908	0.046	0.9600	1.047	1.047	1.24
	351.82	0.933	0.046	0.9820	1.038	1.038	1.20
≈15.0	353.00	0.728	0.067	0.8530	1.106	1.510	1.64
	352.87	0.767	0.067	0.8760	1.083	1.581	1.53
	352.55	0.769	0.065	0.8750	1.092	1.614	1.52
	352.21	0.785	0.064	0.8880	1.100	1.611	1.51
	352.18	0.804	0.064	0.9000	1.089	1.647	1.48
	352.20	0.832	0.068	0.9210	1.076	1.716	1.40
	352.02	0.884	0.059	0.9550	1.058	1.727	1.35
	351.98	0.891	0.068	0.9670	1.064	1.807	1.32

^a Standard uncertainties: $u(T) = \pm 0.5$ K, $u(p) = \pm 0.01$ kPa, $u(x_1) = u(y_1) = \pm 0.001$, T: temperature, x: liquid phase mole fraction, y: vapor phase mole fraction, γ : activity coefficient, α : relative volatility.

Table 3Isobaric VLE data^a at 101.3 kPa for ethanol (1)-water (2)-2HEAB (3).

% $w_{3,initial}$	T (K)	x_1	x_3	y_1	γ_1	γ_2	α_{12}
≈5.1	353.05	0.452	0.011	0.656	1.366	1.349	2.27
	352.50	0.530	0.013	0.683	1.238	1.496	1.86
	351.88	0.626	0.013	0.724	1.138	1.685	1.52
	351.85	0.732	0.014	0.792	1.066	1.799	1.33
	351.58	0.817	0.017	0.855	1.042	1.946	1.19
	351.63	0.872	0.014	0.894	1.019	2.067	1.10
	351.33	0.924	0.015	0.942	1.024	2.137	1.06
≈10.3	351.96	0.619	0.027	0.744	1.179	1.589	1.66
	352.57	0.677	0.029	0.783	1.109	1.582	1.56
	351.54	0.714	0.030	0.797	1.113	1.770	1.41
	351.70	0.772	0.030	0.837	1.074	1.826	1.32
	351.63	0.829	0.032	0.878	1.052	1.951	1.21
	351.53	0.867	0.032	0.910	1.047	1.988	1.18
	351.65	0.911	0.033	0.950	1.035	2.018	1.15
	351.35	0.939	0.033	0.975	1.043	2.079	1.12
≈15.6	352.05	0.556	0.041	0.723	1.271	1.505	1.89
	352.15	0.650	0.043	0.779	1.167	1.569	1.67
	351.75	0.752	0.045	0.842	1.107	1.731	1.44
	351.83	0.792	0.051	0.873	1.087	1.784	1.36
	351.75	0.840	0.052	0.910	1.071	1.843	1.30
	351.61	0.892	0.050	0.951	1.060	1.878	1.25
	351.60	0.915	0.055	0.972	1.057	2.008	1.15

^a Standard uncertainties: $u(T) = \pm 0.5$ K, $u(p) = \pm 0.01$ kPa, $u(x_1) = u(y_1) = \pm 0.001$, T: temperature, x: liquid phase mole fraction, y: vapor phase mole fraction, γ : activity coefficient, α : relative volatility.

Conformation plays an important role in COSMO-RS results since different conformers of the same molecule can have different energy, polarity and hydrogen bonding capacity. It is also worth mentioning, that the conformer with the lowest energy does not necessarily yields the best phase equilibrium results. Paduszynski (Paduszynski (2017)) have studied the effect of ethanol conformers in its σ -profile and found differences in screening charge distribution that has a noticeable impact on the predicted values of activity coefficient at infinite dilution of ethanol. On these grounds, a conformational analysis has been carried out for ethanol and isopropanol molecules in order to find the conformers of alcohols that lead to a better description of VLE of the aqueous mixtures

studied in this work. A number of conformers have been studied and the ones that gave the best results have been used for further analysis. VLE predictions of ethanol-water and isopropanol-water mixtures using the conformers from the COSMOtherm database and the ones determined in this work are presented in Fig. 3, in terms of relative volatilities versus the liquid molar composition of the alcohol. The improvement obtained with the new conformers is significant, especially at the azeotropic point. In order to evaluate the new conformers presented here, VLE predictions of aqueous mixture of alcohols have been performed, using the new conformers developed in this work and the ones of COSMOtherm database. The results are compared against experimental data in

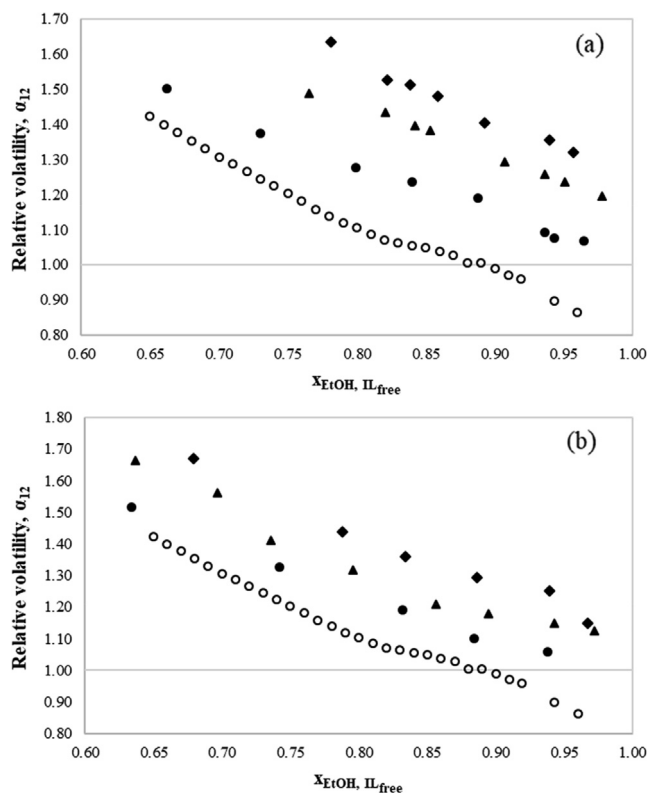


Fig. 1. Experimental relative volatilities of the (a) ethanol (1) – water (2) – 2HEAF (3) system: (●) $\approx 5.2\%$, (▲) $\approx 10.3\%$ and (◆) $\approx 15.0\%$ mass fraction IL and (b) ethanol (1) – water (2) – 2HEAB (3) system: (●) $\approx 5.1\%$, (▲) $\approx 10.3\%$ and (◆) $\approx 15.6\%$ mass fraction. The open circles (○) correspond to the binary ethanol-water (Gmehling, 1977).

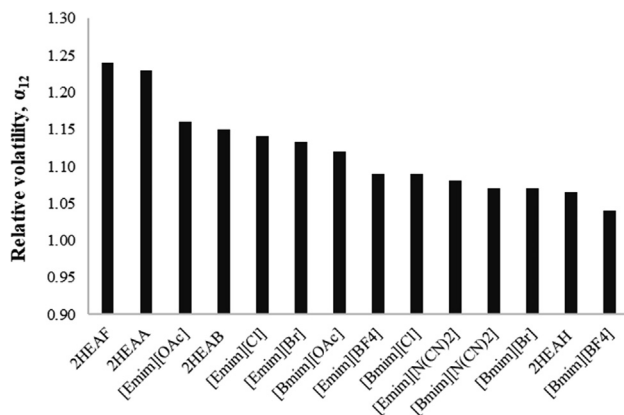


Fig. 2. Effect of ILs on the relative volatility for the ethanol (1) – water (2) system at $x_{\text{ethanol}} = 0.95$ and $w_{\text{IL}} = 0.1$. VLE experimental data for [Emim][OAc], [Bmim][OAc], [Emim][Cl], [Bmim][Cl], [Bmim][BF₄], [Emim][BF₄], [Emim][N(CN)₂], [Bmim][N(CN)₂] were taken from (Ge et al., 2008), [Bmim][Br] from (Zhao et al., 2006), [Emim][Br] from (Tsanas et al., 2014), 2HEAA and 2HEAH from our previous work (Gjineci et al., 2016) and 2HEAF and 2HEAB from this work.

Figs. S3 and S4 of the supplementary material. Furthermore, excess enthalpies of the previously mentioned mixtures have been also investigated and the results are also presented in supplementary material (Figs. S5 and S6). In addition to this, VLE prediction for binary mixtures of ethanol and isopropanol with different polar and non-polar compounds are presented in Figs. S6–S9. The results demonstrate that aqueous mixtures of alcohols are better described with the new conformer in all cases.

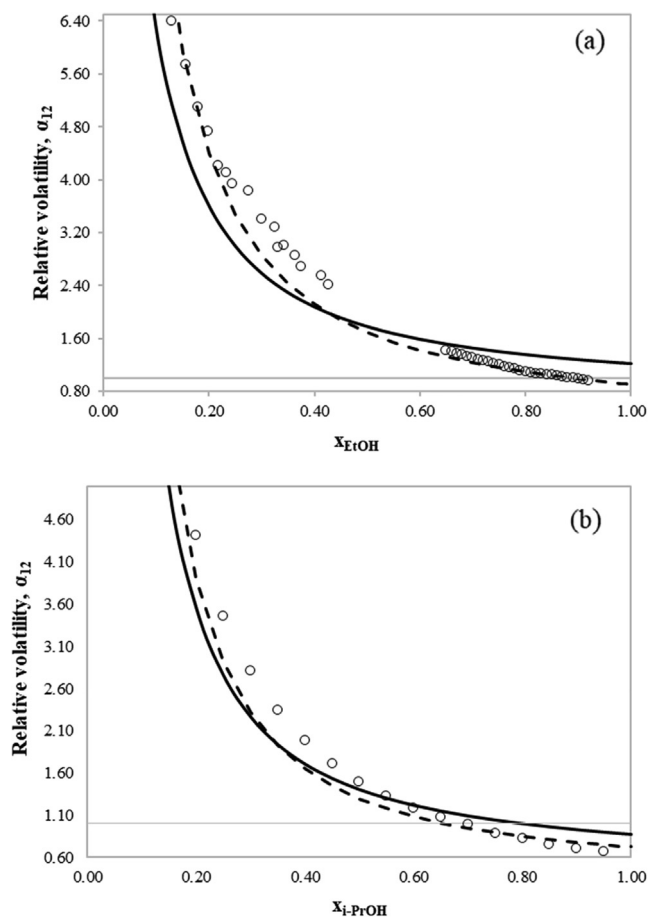


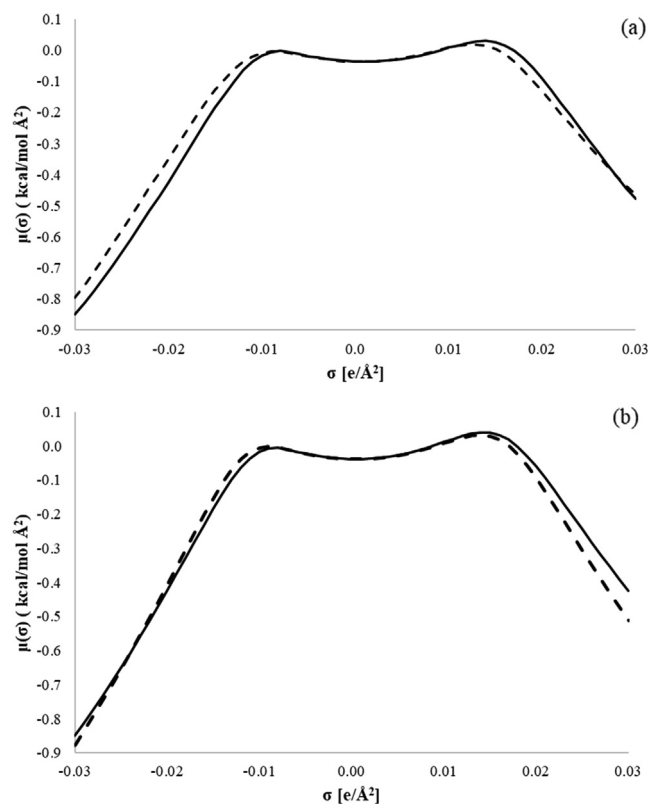
Fig. 3. Experimental (○) and predicted relative volatilities for the binary (a) ethanol (1) – water (2) system and (b) isopropanol (1) – water (2) system. The solid curve corresponds to the ethanol conformer of COSMOtherm database and dashed curve to the new conformer. Experimental data used were taken from (Gmehling, 1977).

Concerning excess enthalpies, COSMO-RS predictions for both conformations of ethanol and isopropanol show noticeable deviations from the experimental data. The model is not able to describe neither qualitatively nor quantitatively the experimental excess enthalpies of the mixtures. Nevertheless, the predicted excess enthalpies with the new conformers are closer to the experimental data than those predicted with the conformers of the COSMOtherm database. On the other hand, when it comes to binary mixtures of alcohols with non-polar compounds the alcohols' conformers of COSMOtherm database show better agreement with the experimental data. In the case of polar-polar binary systems, no significant differences between the conformers were found. As shown in Table 4, the energy differences between the conformers are insignificant. Polarity and hydrogen bonding capacity of the two conformers are reported in Table 4 in terms of COSMO sigma-moments, which are molecular descriptors derived from COSMO-RS calculations. Among the five COSMO sigma-moments available, the second sigma moment (sig2) and the hydrogen bond moments (HB_acc3 and HB_don3), are chemically corresponding to the measures of polarity, hydrogen bond acceptor and donor capacity, respectively. Based on sig2 values, the ethanol conformer of the COSMOtherm database is described as a more polar compound and stronger hb donor and acceptor than the new conformer. On the other hand, the new conformer of isopropanol appears to be a less polar compound and a stronger hb donor as compared to that of COSMOtherm database. These findings come in line with the σ -potential of the compounds shown in Fig. 4 for ethanol and

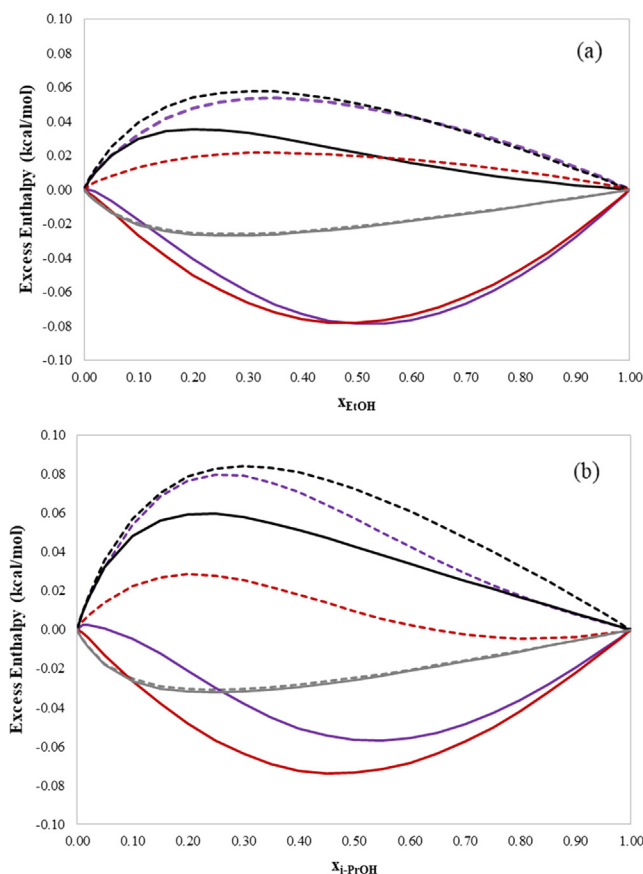
Table 4

Sigma moments for ethanol and isopropanol conformers.

	Molecule			
	Ethanol		Isopropanol	
	Database	This work	Database	This work
sig2	51.5400	46.7716	50.0964	51.4984
Hb_acc3	4.0660	3.0629	3.9418	3.4609
Hb_don3	1.6401	1.3653	1.2792	1.6181

**Fig. 4.** σ -potential of (a) ethanol and (b) isopropanol. Solid curves correspond to the conformers of COSMOtherm database and dashed curves to the conformer of this work.

isopropanol. Apparently, the ethanol conformer of this work seems to be less attractive to hb-donors due to the greater σ -potential at $\sigma < -10^{-2}$ [e/Å²], while isopropanol conformer of this work is more attractive to hb-acceptors due to the lower σ -potential at $\sigma > 10^{-2}$ [e/Å²]. The energy analysis of the mixtures provides a thorough understanding of the interactions between water and alcohols. The calculated excess enthalpies of the mixtures have been taken into consideration in order to evaluate the type of interactions determining the behavior of mixtures. In COSMO-RS, the excess enthalpy of a mixture is calculated as the algebraic sum of three contributions associated with the following intermolecular interactions: hydrogen bonding, van der Waals and misfit interactions. Fig. 5a shows that the main differences between two conformers of ethanol appear in hydrogen bonding and misfit energy. The dominant interaction energy in the case of ethanol conformer of COSMOtherm database between ethanol and water is the hydrogen bonding, while in the case of the new conformer, the misfit energy. For isopropanol, the same conclusions can be drawn as depicted in Fig. 5b. In both cases, conformer of alcohols of COSMOtherm database gives strong negative hb interactions. On the basis of these results, it is concluded that COSMO-RS model overpredicts the hydrogen bond energies.

**Fig. 5.** Excess energy contributions (misfit: black lines, hydrogen bonding: red lines and van der Waals: grey lines, total: purple lines) to the total excess enthalpy of (a) ethanol-water and (b) isopropanol-water mixture. Solid curves correspond to the conformer of COSMOtherm database and dashed curves to the conformer of this work.

5.2.2. VLE prediction of ternary alcohol-water-ILs mixtures

The phase behavior of the ethanol-water and isopropanol-water mixtures with 2HEAF, 2HEAA and 2HEAB has been predicted using the two different approaches for treating ILs as discussed in Section 4, i.e. [C + A] and [CA]. The experimental data for the isopropanol-water-ILs mixtures and for ethanol-water-2HEAA have been taken from our previous work (Gjineci et al., 2016). The effect of ILs anions as well as their initial concentration in the mixture have been taken into account. Experimental relative volatilities of all mixtures along with COSMO-RS predictions are shown in Figs. 6 and 7, with respect to the liquid molar composition of alcohols in IL free-basis.

The results indicate that the ion-pair as well as the distinct ion approach provide qualitative correct VLE predictions of all the systems studied, predicting accurately the effect of the IL anion and IL concentration on relative volatility. The ion-pair approach yields much better quantitative results than the distinct ion with small

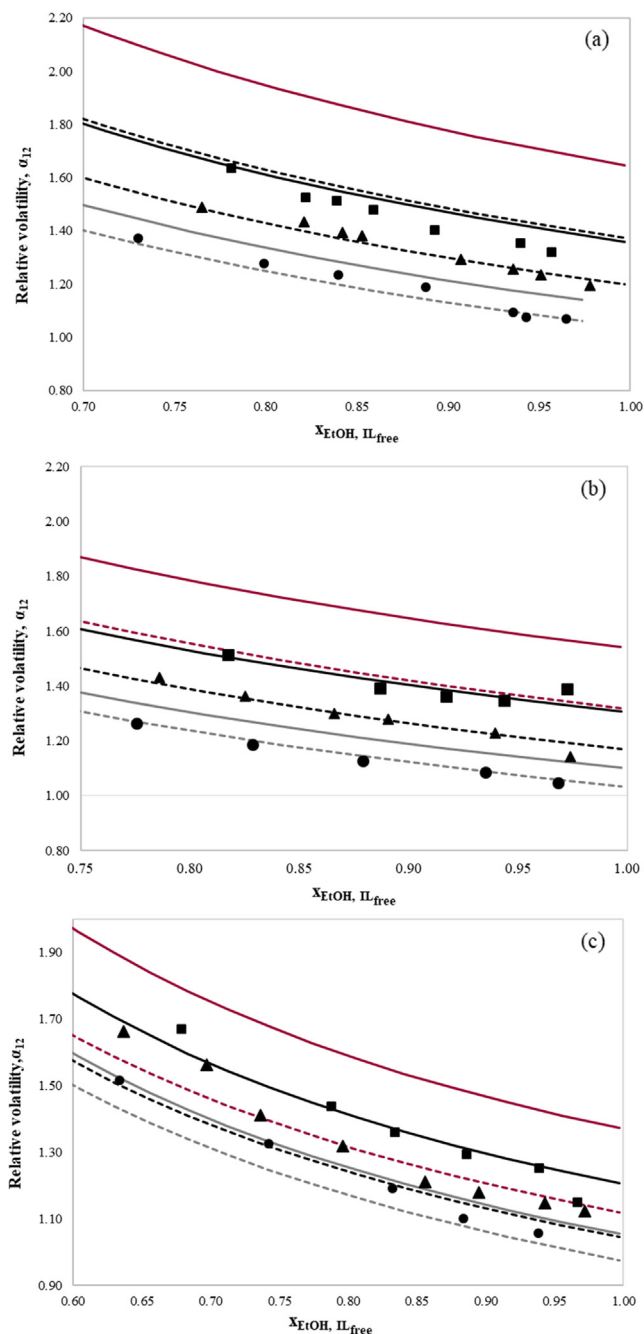


Fig. 6. Experimental and predicted relative volatilities of the (a) ethanol (1) – water (2) – 2HEAF (3) (b) ethanol (1) – water (2) – 2HEAA (3) and (c) ethanol (1) – water (2) – 2HEAB (3) system at 5% mass fraction IL (grey lines), 10% mass fraction IL (black lines) and 15% mass fraction IL (red lines); solid curves correspond to [C + A] model and dashed curves to [CA] model predictions. The experimental ethanol–water–2HEAA VLE data has been taken from our previous work (Gjineci et al., 2016).

deviations from the experimental data. The σ -profiles and σ -potentials of the 2HEAF calculated by the two approaches are shown in Fig. 8. For 2HEAA and 2HEAB these data can be found in the supplementary material (Fig. S10). From the σ -profiles it is concluded that the [C + A] approach has higher peaks in both H-bond acceptor and H-bond donor region, while the [CA] model predicts a higher charge concentration of the molecular surface within the nonpolar region. Hence, the [C + A] model describes the IL as a more polarized compound than the [CA] model, which includes the effect of the cation – anion interactions on the IL charge distribu-

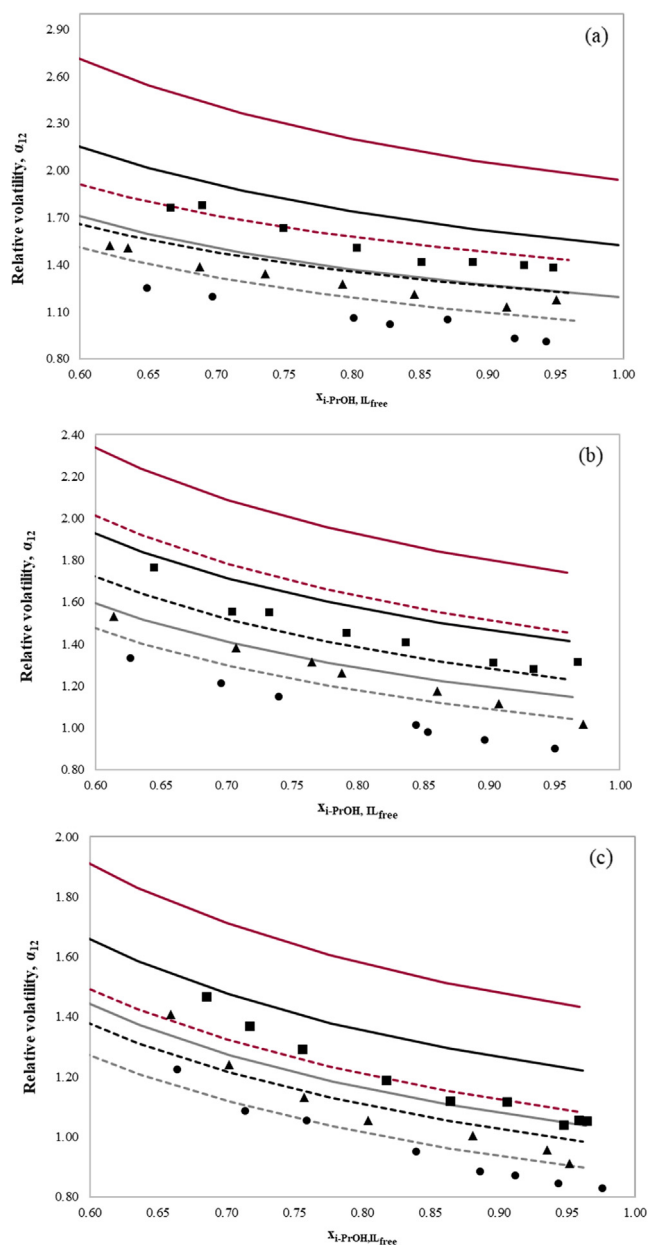


Fig. 7. Experimental and predicted relative volatilities of the (a) isopropanol (1) – water (2) – 2HEAF (3) (b) isopropanol (1) – water (2) – 2HEAA (3) and (c) isopropanol (1) – water (2) – 2HEAB (3) system at 10% mass fraction IL (grey lines), 15% mass fraction IL (black lines) and 20% mass fraction IL (red lines); solid curves correspond to [C + A] model and dashed curves to [CA] model predictions. Experimental data for all the systems were taken from our previous work (Boli et al., 2018).

tion. The higher ionic nature of the IL related to the [C + A] model might explain the overestimation of the electrostatic interactions between the species in solution comparatively to the [CA] model. These observations, are also confirmed by inspecting the σ -potential of ILs. The [C + A] model describes the ILs to be more more attractive for hb-donors and hb-acceptors, which is expressed by lower σ -potentials in the hydrogen bonding region than the [CA] model.

For a better understanding of the molecular interactions between the components of mixtures, the interaction energy values in terms of the different intermolecular interactions, i.e. hydrogen bonding, van der Waals forces and electrostatic interactions, have been analyzed. Their energetic contribution to the total energy

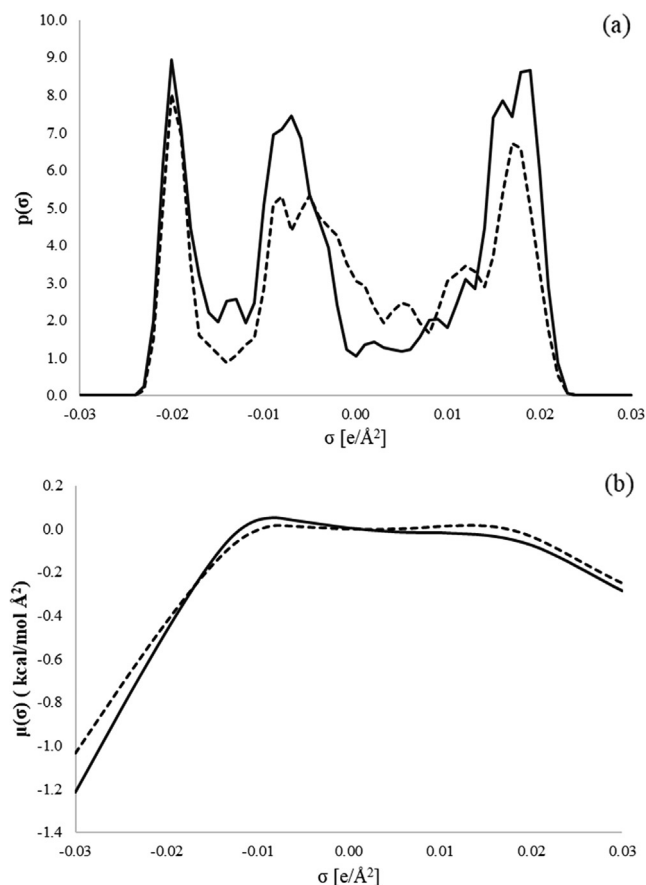


Fig. 8. (a) σ -profile and (b) σ -potential of 2HEAF; solid curves correspond to [C + A] model and dashed curves to [CA] model predictions.

of ethanol-water-ILs mixture and isopropanol-water-ILs mixture was calculated using both [CA] and [C + A] approaches, and the results are presented in Fig. 9. It is shown that hydrogen bonding is the dominant interaction energy in all studied mixtures, van der Waals forces follow, while misfit interactions have the smallest contribution. According to the two different approaches, the [C + A] approach predicts almost twice hb energy values as compared to [CA], which is consistent with our previous observations based on σ -profiles and σ -potentials.

6. Conclusions

This work provides a comprehensive study of the thermodynamic behavior of mixtures containing alcohols, water and ILs by combining isobaric vapor-liquid experimental measurements with quantum-chemical calculations. The isobaric VLE data for the ternary system ethanol-water-ILs indicate that 2HEAF and 2HEAB ILs are promising entrainers for the extraction of ethanol from its aqueous solutions. In the presence of these two ILs the relative volatility of the ethanol-water mixture increases, facilitating an easier separation of the azeotropic mixture. COSMO-RS has been used as a suitable computational method to predict the phase behavior of mixtures containing alcohols, water and ILs. The results show that COSMO-RS predictions are in good qualitative agreement with the experimental data, but the quantitative results should be considered carefully. It should be noticed that a significant contribution of hydrogen bond interactions in the behavior of the mixtures studied is observed. Furthermore, it has been found that the representation of ILs in COSMO-RS using the ion-pair approach, leads to the most accurate predictions. In conclusion,

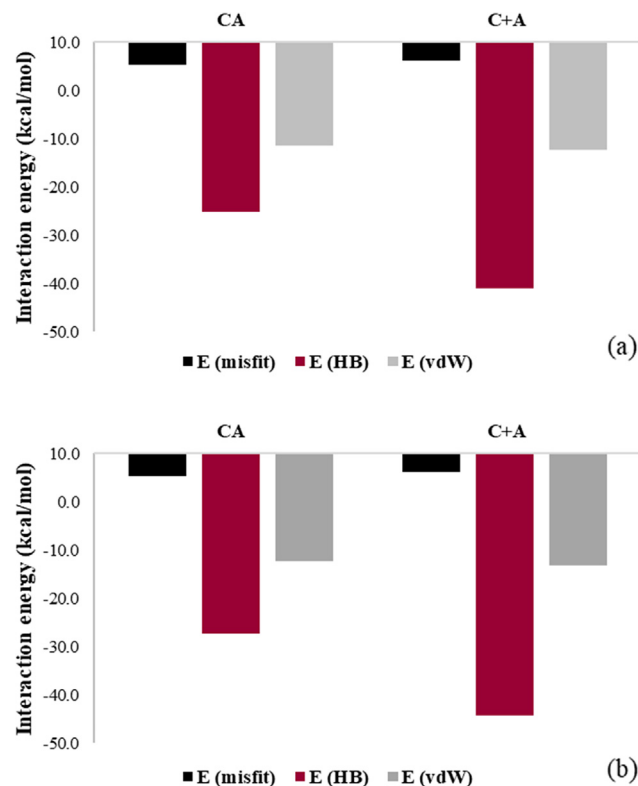


Fig. 9. Misfit, hydrogen bonding (HB) and van der Waals (vdW) energetic contributions in the (a) ethanol-water-2HEAF mixture at 352 K and (b) isopropanol-water-2HEAF mixture at 355 K.

the obtained results are encouraging and implicate that COSMO-RS may be a valuable tool for the a priori prediction of the thermodynamic behavior of ILs with common solvents, allowing to greatly reduce the experimental efforts of finding or designing appropriate ILs for specific applications.

CRedit authorship contribution statement

Elenitsa Boli: Conceptualization, Investigation, Software, Validation, Writing - review & editing. **Epaminondas Voutsas:** Conceptualization, Investigation, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2020.115579>.

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